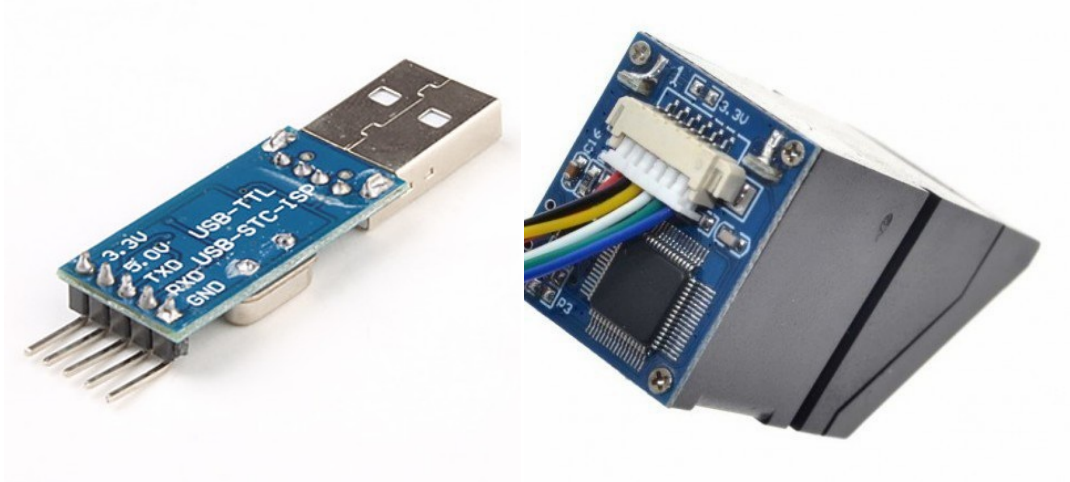


Aim: Interface fingerprint module with raspberry pi

Fingerprint module pins	Color	Wire	USB-TTL convertor
1: VCC	Red	Red	Connect it to USB-TTL(5V) : VCC
2: GND	Black	Black	Connect it to USB-TTL: GND
3: Tx	Yellow	Yellow	Connect it to USB-TTL: Rx
4: Rx	White	White	Connect it to USB-TTL: Tx



#pip3 install pyfingerprint

Create a new file finger.py in pyfingerprint directory

```
pi@raspberrypi: ~/pyfingerprint/src/files/pyfingerprint
File Edit Tabs Help
pi@raspberrypi:~/pyfingerprint/src/files/pyfingerprint $ sudo nano fingerp.py_
```

#fingerprint program starts

import time

from pyfingerprint. pyfingerprint import PyFingerprint

import RPi.GPIO as gpio

print("Successfully imported fingerprint module")

```
gpio.setwarnings(False)
gpio.setmode(gpio.BCM)
try:
    f = PyFingerprint('/dev/ttyUSB0', 57600, 0xFFFFFFFF, 0X00000000)
except Exception as e:
    print('Exception Message :' + str(e))
```

```
def enrollFinger():
    print ('Enrolling Finger')
    time.sleep(3)
    print('Place Finger')
    while(f.readImage()==False):
        pass
    f.convertImage(0X01)
    result=f.searchTemplate()
    positionNumber = result[0]
    if (positionNumber >=0):
        print('Template already exists at position #' +str(positionNumber))
        time.sleep(2)
        return
    print('Remove finger')
    print('Waiting for same finger')
    time.sleep(3)
    while(f.readImage()==False):
        pass
    f.convertImage(0X02)
    if(f.compareCharacteristics()==0):
        print('Fingers do not match')
        time.sleep(2)
        return
    else:
        f.createTemplate()
        positionNumber=f.storeTemplate()
        print('Finger enrolled successfully')
        print('Stored at pos: ' +str(positionNumber))
```

```
        time.sleep(2)
def searchFinger():
    try:
        print('Waiting for finger...')
        time.sleep(2)
        while(f.readImage()==False):
            time.sleep(4)
        return
        f.convertImage(0X01)
        result = f.searchTemplate()
        positionNumber = result[0]
        if positionNumber==-1:
            print('No Match found')
            time.sleep(4)
            return False
        else:
            print('Found template at position ' + str(positionNumber))
            time.sleep(4)
            return True
    except Exception as e:
        print('Operation failed')
        print('Exception message: ' +str(e))
        exit(1)
    return
```

```
def deleteFinger():
    try:
        print('Waiting for finger...')
        time.sleep(2)
        while(f.readImage()==False):
            time.sleep(4)
        return
        f.convertImage(0X01)
        result = f.searchTemplate()
        positionNumber = result[0]
```

```

        if positionNumber==-1:
            print('No Match found')
            time.sleep(4)
            return False
        else:
            if(f.deleteTemplate(positionNumber)==True):
                print('Finger Deleted')

                time.sleep(4)
                return True
    except Exception as e:
        print('Operation failed')
        print('Exception message: ' +str(e))
        exit(1)
    return

time.sleep(1)

print('Start')
while True:
    i = int(input("\nEnter : \n1.Enroll\n2.Search\n3.Delete\n4.Exit\n"));
    if i==1:
        enrollFinger()
    elif i==2:
        searchFinger()
    elif i==3:
        deleteFinger()
    elif i==4:
        break
    else:
        print("Invalid option !!!Enter Correct")
#fingerprint program ends

```

